

Problem Set - 2

b-) Density matrix fundamental

a) given

$$\rho = \begin{pmatrix} \frac{1}{4} & n \\ n^* & p \end{pmatrix}$$

Required

$$p = ?$$

$$n = ?$$

Solution

By using fundamental properties

$$\text{Tr}(\rho) = 1$$

$$\text{Tr} \begin{pmatrix} \frac{1}{4} & n \\ n^* & p \end{pmatrix} = 1$$

$$p + \frac{1}{4} = 1$$

$$p = 1 - \frac{1}{4}$$

$$p = \frac{4-1}{4}$$

$$p = \frac{3}{4}$$

By using the density property

$$\text{Tr}(\rho^2) = 1 \text{ For pure state}$$

$$\text{Tr}(\rho^2) = 1$$

$$\text{Tr} \left[\begin{pmatrix} \frac{1}{4} & n \\ n^* & \frac{3}{4} \end{pmatrix} \begin{pmatrix} \frac{1}{4} & n \\ n^* & \frac{3}{4} \end{pmatrix} \right] = 1$$

$$\text{Tr} \begin{bmatrix} \frac{1}{16} + nn^* & \frac{n}{4} + \frac{3n}{4} \\ \frac{1}{4}n^* + \frac{3}{4}n^* & n^*n + \frac{9}{16} \end{bmatrix} = 1$$

$$\text{Tr} \begin{bmatrix} \frac{1}{16} + |n|^2 & \frac{4n}{4} \\ \frac{4n^*}{4} & |n|^2 + \frac{9}{16} \end{bmatrix} = 1$$

$$\frac{1}{16} + |n|^2 + |n|^2 + \frac{9}{16} = 1$$

$$2|n|^2 + \frac{10}{16} = 1$$

$$2|n|^2 = 1 - \frac{10}{16}$$

$$2|n|^2 = \frac{16-10}{16}$$

$$2|n|^2 = \frac{6}{16}$$

$$|n|^2 = \frac{3}{16}$$

$$|n| = \frac{\sqrt{3}}{4}$$

$$n = \frac{\sqrt{3}}{4}$$

①

b) Given

$$S = \sum_i p_i S_i$$

Required

$$1 - S^\dagger = S$$

$$2 - \text{Tr}(S) = 1$$

$$3 - \langle \psi | S | \psi \rangle \geq 0$$

Show All these

Solution

1) Hermiticity, $S^\dagger = S$

$$S^\dagger = (S)^\dagger$$

$$S^\dagger = \left(\sum_i p_i S_i \right)^\dagger \\ = \sum_i p_i^* S_i^\dagger$$

Where, $\boxed{S_i^\dagger = S_i}$

$$S^\dagger = \sum_i p_i^* S_i, \quad \underbrace{p_i^* = p_i}_{\text{Because of Real value}}$$

$$S^\dagger = \sum_i \underbrace{p_i}_{=1} S_i \\ = S$$

$$\underline{S^\dagger = S} \quad \checkmark$$

2. Trace Condition

$$\text{Tr}(S) = 1$$

$$\text{Tr}\left(\sum_i p_i S_i\right) = 1$$

$$\sum_i \underbrace{p_i}_{=1} \text{Tr}(S_i) = 1$$

$$1 \times 1 = 1 \Rightarrow 1 = 1 \quad \underline{\underline{\checkmark}}$$

$$\rightarrow \text{Tr}(S) = 1 \\ 1 = 1 \quad \checkmark$$

3. Positivity

$$\langle \psi | S | \psi \rangle \geq 0$$

$$\langle \psi | \sum_i p_i S_i | \psi \rangle \geq 0$$

$$\sum_i \underbrace{p_i}_{=1} \langle \psi | S_i | \psi \rangle \geq 0$$

$$1 \times \langle \psi | S_i | \psi \rangle \geq 0$$

$$\underline{\underline{\langle \psi | S | \psi \rangle \geq 0}}$$

c) Positive Semi-definite (non-negative)

Given

$$\langle \psi | A | \psi \rangle \geq 0$$

Required

$\lambda \geq 0$ - Show this

Solution

Let's say λ is eigenvalue of matrix A

$$A|\lambda\rangle = \lambda|\lambda\rangle$$

Therefore

$$\langle \lambda | A | \lambda \rangle \geq 0 \Rightarrow \langle \lambda | \underbrace{A | \lambda \rangle}_{\lambda | \lambda \rangle} \geq 0,$$

$$\langle \lambda | \lambda \rangle \lambda \geq 0 \Rightarrow \lambda \underbrace{\langle \lambda | \lambda \rangle}_{=1} \geq 0$$

$$\lambda \geq 0$$

$$\underline{\lambda \geq 0}$$

d) Given

$$\delta = \frac{1}{2}(I + a \cdot \sigma)$$

Required

$$\text{Tr}(\delta^2) = \frac{1}{2}(1 + a \cdot a)$$

Solution

$$\begin{aligned} \text{Tr}(\delta^2) &= \text{Tr}\left(\left(\frac{1}{2}(I + a \cdot \sigma)\right)^2\right) = \text{Tr}\left(\frac{1}{4}(I^2 + I a \cdot \sigma + a \cdot \sigma I + (a \cdot \sigma)(a \cdot \sigma))\right) \\ &= \text{Tr}\left(\frac{1}{4}(I^2 + 2I a \cdot \sigma + a \cdot a \sigma^2)\right) \end{aligned}$$

$$= \text{Tr}\left(\frac{1}{4}(I^2 + 2a \cdot \sigma + a \cdot a I)\right), \text{ where}$$

$$= \frac{1}{4}(\text{Tr}(I^2) + 2a \text{Tr}(\sigma) + a \cdot a \text{Tr}(I))$$

$\sigma^2 = I$ $I^2 = I$ $2I a \cdot \sigma = 2a \cdot \sigma$
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$$\text{where } \left. \begin{array}{l} \text{Tr}(I) = 2 \\ \text{Tr}(\sigma) = 0 \end{array} \right\} \text{ Always true}$$

(3)

$$\begin{aligned}
 \text{Tr}(S^2) &= \frac{1}{4} (2 + 2a \cdot 0 + a \cdot a - 2) \\
 &= \frac{1}{4} (2 + 2a \cdot a) \\
 &= \frac{2}{4} (1 + a \cdot a) \\
 &= \frac{1}{2} (1 + a \cdot a)
 \end{aligned}$$

2)

a) Given

$$S = \sum_i^d |P\rangle\langle P|$$

Required

$$S = \sum_i^d \frac{1}{d} |P\rangle\langle P| - \text{show this}$$

Solution

$$\begin{aligned}
 \text{Tr}(S^2) &= \text{Tr} \left(\left(\sum_i^d |P\rangle\langle P| \right)^2 \right) \quad \text{where } \underbrace{(|P\rangle\langle P|)^2 = |P\rangle\langle P|} \\
 &= \text{Tr} \left(\sum_i^d |P\rangle\langle P| \right)
 \end{aligned}$$

Using Lagrange Multiplier

To minimize f subject to $g \rightarrow$ the Lagrange function is

$$\begin{aligned}
 L(\lambda_1, \dots, \lambda_d, u) &= f(\lambda_1, \dots, \lambda_d) - y(g(\lambda_1, \dots, \lambda_d) - 1) \\
 &= \sum_{i=1}^d \lambda_i^2 - y \left(\sum_{i=1}^d \lambda_i^2 - 1 \right)
 \end{aligned}$$

Derivate with respect of λ_i

$$\frac{\partial L}{\partial \lambda_i} = 0$$

$$\frac{\partial}{\partial \lambda_i} \left(\sum_{i=1}^d \lambda_i^2 - y \left(\sum_{i=1}^d \lambda_i^2 - 1 \right) \right)$$

(4)

$$\frac{\partial}{\partial \lambda_i} \left(\sum_{i=1}^d \lambda_i^2 - \gamma \left(\sum_{i=1}^d \lambda_i - 1 \right) \right) = 0$$

$$2\lambda_i - \gamma = 0$$

$$\boxed{\lambda_i = \frac{\gamma}{2}}, \text{ where } i = 1, 2, \dots, d$$

Derivate with respect of γ

$$\frac{\partial L}{\partial \gamma} = 0$$

$$\frac{\partial}{\partial \gamma} \left(\sum_{i=1}^d \lambda_i^2 - \gamma \left(\sum_{i=1}^d \lambda_i - 1 \right) \right) = 0$$

$$- \left(\sum_{i=1}^d \lambda_i - 1 \right) = 0$$

$$- \sum_{i=1}^d \lambda_i + 1 = 0$$

$$- \sum_{i=1}^d \lambda_i = -1 \Rightarrow \sum_{i=1}^d \lambda_i = 1$$

$$\text{, where } \boxed{\sum_{i=1}^d 1 = d}$$

$$d\lambda_i = 1$$

$$\underline{\underline{\lambda_i = \frac{1}{d}}}$$

$$S_{\text{min}} = \sum_{i=1}^d \lambda_i |i\rangle\langle i| = \sum_{i=1}^d \frac{1}{d} |i\rangle\langle i|$$

$$= \frac{1}{d} \sum_{i=1}^d |i\rangle\langle i|$$

$$S_{\min} = \frac{1}{d} \underbrace{\sum_{i=1}^d |p_i\rangle\langle i|}_{\mathbf{I}}$$

$$\underline{\underline{S_{\min} = \frac{\mathbf{I}}{d}}}$$

Calculating the minimal purity

$$\text{Tr}(S_{\min}^2) = \text{Tr}\left(\sum_{i=1}^d \lambda_i^2 (|i\rangle\langle i|)^2\right)$$

$$= \text{Tr}\left(\sum_{i=1}^d \lambda_i^2 |i\rangle\langle i| |i\rangle\langle i|\right)$$

$$= \sum_{i=1}^d \lambda_i^2 \text{Tr}\left(|i\rangle\langle i| \underbrace{|i\rangle\langle i|}_{\mathbf{I}}\right)$$

$$= \sum_{i=1}^d \lambda_i^2 \text{Tr}(|i\rangle\langle i|) \text{ where, } \text{Tr}(|i\rangle\langle i|) = \underbrace{\langle i|i\rangle}_{=1}$$

$$\underline{\underline{\text{Tr}(|i\rangle\langle i|) = 1}}$$

$$\text{Tr}(S_{\min}^2) = \sum_{i=1}^d \lambda_i^2 \cdot 1$$

$$= \sum_{i=1}^d \lambda_i^2 \text{ where } \left. \begin{array}{l} \sum_{i=1}^d = d \\ \lambda_i = \frac{1}{d} \end{array} \right\} \text{Substituted}$$

$$\text{Tr}(S_{\min}^2) = d \cdot \left(\frac{1}{d}\right)^2$$

$$= d \cdot \frac{1}{d^2} = \frac{1}{d}$$

$$\underline{\underline{\text{Tr}(S_{\min}^2) = \frac{1}{d}}}$$

(b) Given

$$\text{Tr}(AB) = \text{Tr}(BA)$$

Required

1- $\text{Tr}([A, B]) = 0$ - Show this

2- ~~Subsequent~~

2- $\text{Tr}(S^k) = \text{Tr}(S)$ - Show this

Solution

1, $\langle p | A | 0 \rangle$

$$B_{ij} = \langle j | B | i \rangle$$

$$\text{Tr}(AB) = \sum_{i=1}^d \langle i | AB | i \rangle$$

where

$$\text{Tr}(BA) = \sum_{i=1}^d \langle i | BA | i \rangle$$

$$\text{Tr}(BA) = \sum_{i,j} B_{ji} A_{ij}$$

where, $I = \sum_j |j\rangle\langle j|$

$$\text{Tr}(AB) = \sum_{i=1}^d \sum_{j=1}^d \langle i | A | 0 \rangle \langle j | B | i \rangle = \sum_{i,j} A_{ij} B_{ji}$$

$$\text{Tr}(AB) = \sum_{i,j} A_{ij} B_{ji}, \quad \text{Tr}(BA) = \sum_{i,j} B_{ji} A_{ij}$$

The two sum are identical

$$\boxed{\sum_{i,j} A_{ij} B_{ji} = \sum_{i,j} B_{ji} A_{ij}}$$



$$\text{Tr}(AB) = \text{Tr}(BA)$$

Trace of commutator is vanished

$$[A, B] = AB - BA$$

$$\text{Tr}([A, B]) = \text{Tr}(AB - BA)$$

$$\text{Tr}([A, B]) = \text{Tr}(AB) - \text{Tr}(BA) \\ = \underline{\underline{0}}$$

Conservation of normalization under unitary dynamics

Unitary transformation U to a new state S'

$$S \xrightarrow{U} S'$$

$$S' = U S U^\dagger$$

Normalization is conserved i.e.

$$\text{Tr}(S') = \text{Tr}(S)$$

$$\text{Tr}(S') = \text{Tr}(U S U^\dagger)$$

The cyclic property of the trace

$$\boxed{\text{Tr}(XYZ) = \text{Tr}(ZXY)}$$

~~where~~
$$\text{Tr}(S') = \text{Tr}(U S U^\dagger) \\ = \text{Tr}(U S U^\dagger)$$

where, $\text{Tr}(U S U^\dagger) = \text{Tr}(S U^\dagger U)$ -> cyclic property

$$\boxed{\text{Tr}(U S U^\dagger) = \text{Tr}(S U^\dagger U)} \quad \dots \dots \dots \text{a x.}$$

$$\text{Tr}(S') = \text{Tr}(S \underbrace{U^\dagger U}_I)$$

$$\text{where } \boxed{U^\dagger U = U U^\dagger = I}$$

Unitary operation

$$\underline{\underline{\text{Tr}(S') = \text{Tr}(S)}}$$

Counter example in infinite dimension Hilbert space

$$\text{Tr}(AB) = \text{Tr}(BA)$$

Example: Position and momentum

$$\begin{aligned}[x, p] &= i\hbar \\ &= i\hbar I\end{aligned}$$

$$\text{Tr}([x, p]) = 0 \Rightarrow \text{Tr}(i\hbar I) = 0$$

But, trace of identity matrix is equal to the dimension of matrix

$$\boxed{\text{Tr}(I) = d}$$

$$\text{Tr}(I) = \int \langle x | I | x \rangle dx = \int \langle x | x \rangle dx = \int dx$$

$$\text{Tr}(I) = \int dx$$

$$\text{Tr}(i\hbar I) = i\hbar \underbrace{\text{Tr}(I)}_{\int dx} = i\hbar \int dx$$

$$\text{Tr}(i\hbar I) = i\hbar \int_{-\infty}^{\infty} dx$$

$$\text{Tr}(i\hbar I) = i\hbar \int_{-\infty}^{\infty} dx$$

$$\underline{\underline{\text{Tr}(i\hbar I) = \infty}}$$

$$\text{Tr}([x, p]) = 0 \text{ and } 0 \neq \infty$$

The trace of the commutator does not vanish

NB: The cyclic property of the trace fails because unbounded operators like position (x) and momentum (p) are not trace class in infinite dimension

c) Show that the reduced density matrix of entangled state is mixed?

Given

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle|1\rangle + |1\rangle|0\rangle)$$

Required

ρ_A, ρ_B is reduced matrix is mixed state ??

Solution

$$\rho = (|\psi\rangle\langle\psi|)_{AB} = \frac{1}{\sqrt{2}} (|0\rangle|1\rangle + |1\rangle|0\rangle) \frac{1}{\sqrt{2}} (\langle 0| \langle 1| + \langle 1| \langle 0|)$$

$$\rho_{AB} = \frac{1}{2} (|0\rangle\langle 0| \otimes |1\rangle\langle 1| + |0\rangle\langle 1| \otimes |1\rangle\langle 0| + |1\rangle\langle 0| \otimes |0\rangle\langle 1| + |1\rangle\langle 1| \otimes |0\rangle\langle 0|)$$

The reduced density matrix I calculated on of the two

I apply partial trace (ρ_A)

$$\rho_A = \text{Tr}_B(\rho_{AB}) = \text{Tr}_B \left(\frac{1}{2} (|0\rangle\langle 0| \otimes |1\rangle\langle 1| + |0\rangle\langle 1| \otimes |1\rangle\langle 0| + |1\rangle\langle 0| \otimes |0\rangle\langle 1| + |1\rangle\langle 1| \otimes |0\rangle\langle 0|) \right)$$

$$\begin{aligned} \rho_A &= \frac{1}{2} (\text{Tr}_B(|0\rangle\langle 0| \otimes |1\rangle\langle 1|) + \text{Tr}_B(|0\rangle\langle 1| \otimes |1\rangle\langle 0|) + \text{Tr}_B(|1\rangle\langle 0| \otimes |0\rangle\langle 1|) + \text{Tr}_B(|1\rangle\langle 1| \otimes |0\rangle\langle 0|)) \\ &= \frac{1}{2} \left(\underbrace{\langle 1|1\rangle}_{=1} |0\rangle\langle 0| + \underbrace{\langle 0|1\rangle}_{=0} |0\rangle\langle 1| + \underbrace{\langle 1|0\rangle}_{=0} |1\rangle\langle 0| + \underbrace{\langle 0|0\rangle}_{=1} |1\rangle\langle 1| \right) \end{aligned}$$

$$\rho_A = \frac{1}{2} (|0\rangle\langle 0| + 0 + 0 + |1\rangle\langle 1|)$$

$$\rho_A = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|)$$

To Show ρ_A is mixed state $\text{Tr}(\rho_A^2) < 1$

$$\begin{aligned} \text{Tr}(\rho_A^2) &= \text{Tr} \left(\left(\frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) \right)^2 \right) \\ &= \text{Tr} \left(\frac{1}{4} (|0\rangle\langle 0| \otimes |0\rangle\langle 0| + |0\rangle\langle 1| \otimes |1\rangle\langle 0| + |1\rangle\langle 0| \otimes |0\rangle\langle 1| + |1\rangle\langle 1| \otimes |1\rangle\langle 1|) \right) \\ &= \text{Tr} \left(\frac{1}{4} (|0\rangle\langle 0| + |1\rangle\langle 1|) \right) \\ &= \frac{1}{4} (\text{Tr}(|0\rangle\langle 0|) + \text{Tr}(|1\rangle\langle 1|)) \end{aligned}$$

(20)

$$\text{Tr}(S_A^2) = \frac{1}{4} (1+1)$$

$$= \frac{2}{4}$$

$$= \frac{1}{2}$$

$\text{Tr}(S^2) = 1$ - pure state

$\text{Tr}(S^2) < 1$ - mixed state

~~mixed state~~

Therefore $\frac{1}{2} < 1$

$$\text{Tr}(S_A^2) < 1$$

S_A is mixed state

~~mixed state~~